12/09/2008

LETTER OF RECOMMENDATION
for
Eliana ZOQUE-LOPEZ

Eliana Zoque-Lopez is a graduate student at the University of Chicago who is working under my supervision. She is expected to defend her PhD in the Spring of 2009.

In her Thesis, Eliana studies the structure of the following set

$$Z_{\text{nil}} := \{(x, y, i, j) \in N_n \times N_n \times \mathbb{C}^n \times (\mathbb{C}^n)^* \mid xy - yx = ij\}, \qquad (*)$$

where N_n denotes the variety of nilpotent complex $n \times n$-matrices, and where i stands for a column, resp. j stands for a row, vector.

To put the above problem in context, let me recall that W.L. Gan and myself introduced a few years ago a variety Z that plays a crucial role in representation theory of rational Cherednik algebras. The variety Z is formed by quadruples (x, y, i, j) similar to those in $(*)$ except that the matrices x and y were not required to be nilpotent. We were able to show that the variety Z is reduced, moreover, it is a complete intersection with $n + 1$ irreducible components.

The variety Z_{nil} in $(*)$ may be viewed as a 'nilpotent version' of the variety Z. In her Thesis, Eliana describes the irreducile components of Z_{nil}. More specifically, Eliana defines explicitly $n + 1$ closed irreducible susets of Z_{nil} and she conjectures that these subsets are precisely the irreducible components of Z_{nil}. At the moment, the conjecture has been verified completely for all $n \leq 8$. Eliana is now in the process of working out a few remaining technical details that would yield the proof of the conjecture in full generality. I am confident that she will be able to write down a complete proof of the conjecture within 2-3 months.

Although the varieties Z and Z_{nil} have similar definitions and are expected to have the same number of irreducible components, these varieties

are quite different. Unlike the case of Z, the variety Z_{nil} is *not* equidimensional, in particular, it is not a complete intersection. Thus, the problem studied by Eliana is, in fact, considerably more difficult than the one solved by Gan and myself.

It might be helpful to compare the above problems with a well-known classical problem of studying the structure of a so-called *commuting variety*, formed by commuting pairs of $n \times n$-matrices. The irreducibility of the commuting variety has been established by M. Gerstenhaber in 1960 by fairly elementary means. However, the irreducibility of a 'nilpotent version' of the commuting variety turned out to be a much harder problem that has been solved much later. Specifically, V. Baranovsky proved in 2002 that the set formed by the pairs of commuting *nilpotent* $n \times n$-matrices is irreducible. The argument used by Baranovsky was much less elementary than the one due to Gerstenhaber; it was based in a crucial way on the irreducibility of the Hilbert scheme of points in the plane, a nontrivial result of algebraic geometry.

The techniques used by Eliana in studying the irreducible components of the set $(*)$ is a combination of ideas similar to those exploited by Baranovsky with arguments based on several really complicated brute force computations. Indeed, all the available arsenal of modern algebraic geometry doesn't seem to provide efficient tools for dealing with this type of varieties. It is a long-standing unsolved problem, for instance, to determine whether or not the commuting variety is reduced.

To summarize, Eliana Zoque-Lopez has succeeded in solving a very difficult concrete problem.

Eliana is a very hard working and determined person, so I have no doubts that she will have a successful career in the future. I am happy to strongly recommend Eliana Zoque-Lopez for a junior position at your Department.

Victor Ginzburg
The University of Chicago,
Department of Mathematics